

Unique response pathways are established by allosteric interactions among nuclear hormone receptors

SCIFACT-EVIDENCE-3203590

Heterodimerization is a common paradigm among eukaryotic transcription factors. The 9-cis retinoic acid receptor (RXR) serves as a common heterodimerization partner for several nuclear receptors, including the thyroid hormone receptor (T3R) and retinoic acid receptor (RAR). This raises the question as to whether these complexes possess dual hormonal responsiveness. We devised a strategy to examine the transcriptional properties of each receptor individually or when tethered to a heterodimeric partner. We find that the intrinsic binding properties of RXR are masked in T3R-RXR and RAR-RXR heterodimers. In contrast, RXR is active as a non-DNA-binding cofactor with the NGFI-B/Nurr1 orphan receptors. Heterodimerization of RXR with constitutively active NGFI-B/Nurr1 creates a novel hormone-dependent complex. These findings suggest that allosteric interactions among heterodimers create complexes with unique properties. We suggest that allostery is a critical feature underlying the generation of diversity in hormone response networks.